

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641080321 A

(19) INDIA

(22) Date of filing of Application :30/06/2026

(43) Publication Date : 03/07/2026

(54) Title of the invention : BRAIN-COMPUTER INTERFACE BASED COGNITIVE STATE DETECTION AND ALERT SYSTEM FOR WORKER SAFETY IN HAZARDOUS ENVIRONMENTS

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|-----------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (51) International classification             | :A61B<br>5/374,<br>A61B<br>5/0476,<br>A61B 5/00,<br>G08B<br>21/04,<br>A61B<br>5/372 | (71)Name of Applicant :<br><b>1)J.J. COLLEGE OF ENGINEERING AND TECHNOLOGY</b><br>Address of Applicant :Ammapettai, Poolangulathupatti (PO),<br>Tiruchirappalli, Tamil Nadu, India Tiruchirappalli Tamil Nadu<br>India |
| (31) Priority Document No                     | :NA                                                                                 | (72)Name of Inventor :<br><b>1)Mrs. G. Deepalakshmi</b>                                                                                                                                                                |
| (32) Priority Date                            | :NA                                                                                 | <b>2)Dr. P. Mathiyalagan</b>                                                                                                                                                                                           |
| (33) Name of priority country                 | :NA                                                                                 | <b>3)Dr. L. Jegatha Deborah</b>                                                                                                                                                                                        |
| (86) International Application No             | :                                                                                   | <b>4)S. Nalina</b>                                                                                                                                                                                                     |
| Filing Date                                   | :01/01/1900                                                                         | <b>5)M. Tharani</b>                                                                                                                                                                                                    |
| (87) International Publication No             | : NA                                                                                | <b>6)R. Narmadha</b>                                                                                                                                                                                                   |
| (61) Patent of Addition to Application Number | :NA                                                                                 |                                                                                                                                                                                                                        |
| Filing Date                                   | :NA                                                                                 |                                                                                                                                                                                                                        |
| (62) Divisional to Application Number         | :NA                                                                                 |                                                                                                                                                                                                                        |
| Filing Date                                   | :NA                                                                                 |                                                                                                                                                                                                                        |

(57) Abstract :

Worker safety is essential in hazardous environments because workers may experience panic, stress, fatigue, and distraction, which can increase the risk of workplace accidents. The invention proposes a Brain-Computer Interface (BCI)-based real-time cognitive state detection system. EEG signals are collected using an electrode to monitor brain activity. Brainwave features extracted from the signals are analyzed using a Machine Learning algorithm called Support Vector Machine (SVM). When an unsafe condition is detected, the system generates an alert to warn the workers and improve workplace safety.

No. of Pages : 14 No. of Claims : 6